

CIV102 Bridge Design Report

Team 409

“Bridge Baddies”

Bridge name: Bridge Baddies Shooting for the Stars ★★

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Introduction

The report explains the various design iterations, what led to the final design and the reasoning behind the choices made when doing the project. Our bridge is a small-scale box bridge made to resist Load Case 1 and as much as Load Case 2 as possible. After multiple iterations to Design 0, the final design achieved a theoretical failure load of 1358.759 N, with a total length of 1255mm. Some key alterations to Design 0 included: adding a double top plate, adjusting the height of all the members, varying the height along the bridge, and adding diaphragms where additional support was needed.

Project Background

All loading assumptions, train geometry and loading protocol used in this analysis follow the CIV102 handout. For Load Case 1, the trains' total weight of 400 N is distributed evenly across six axles. Load case 2 "base case" consists of freight cars of equal weight, and a locomotive of weight $\times 1.35$ that of a freight car. Each subsequent passe increases in weight per the prescribed protocol. Assumptions used throughout the analysis include: treating wheel loads as point loads, and neglecting material imperfections and construction defects in theoretical calculations. Though for the final failure load it was taken into account. These assumptions are consistent with typical introductory structural analysis used in CIV102.

Methodology

To develop a design capable of supporting the greatest load and while maintaining an efficient strength to weight ratio, modifications were made to the reference Design 0. Hand calculations and Python code were performed to evaluate internal forces, sectional properties and predict failure loads. The program calculated shear forces and bending moments at each point along the span, provided graphical evidence, identified the controlling maximum shear and maximum moment for each load case and for the various iterations. These values were then to compute bending moments, shear stresses, and factors of safety for each design iteration.

The analysis consisted of computing key cross-sectional properties such as centroidal location (y), second moment of area (I), and first moment of area for a section (Q), for every modified section. These properties were then used to assess bending and shear capacity and expected failure modes.

Following the initial analysis of Design 0, the code was expanded to evaluate a wider range of geometries. From the cross-sectional geometries, plate heights and layouts were systematically modified while tracking their effects on predicted failure loads and FOS. The following FOS values were calculated: $FOS_{tension}$, $FOS_{compression}$, FOS_{max_shear} , FOS_{glue_shear} , $FOS_{side_flange1}$, $FOS_{main_flange2}$, FOS_{web3} , $FOS_{shear_buckling}$. These changes were guided by course notes and previously solved examples.

Several practical and theoretical considerations influenced our development of the final design. The total area provided imposed a major design constraint. For each iteration, material usage diagrams were created to plan how the matboard could be cut. The final plan on how the mathboard is cut is shown below. Any free space was

Figure 2: Graph of the Bending Moment Envelope for Load Case 2

Minimum FOS: 0.596

Failure Load: 254.392 N

Max Moment: 69575.212 at position 640.800

Max Shear: 256.013 at position 1196.000

Iteration #1 Double-Layering the Top Deck

To address the lowest safety factor found for main flange buckling due to compressive flexural stress. The critical buckling stress for the main flange is governed by plate buckling capacity equation:

$\sigma_{crit} = \frac{4\pi^2 E}{12(1-u^2)} \left(\frac{thickness_{top\ deck}}{b} \right)^2$. In order to increase the buckling capacity and therefore increase the safety factor, we doubled the thickness of the top deck, from 1.27mm to 2.54mm by adding another layer of matboard along the entire span. Since this capacity scales with t^2 , it increased the FOS from 0.596 to 3.417. The compressive flexural stress is computed using $\sigma = \frac{My_{top}}{I}$. Doubling the thickness helped increase the I, which decreased the value of the compressive stress, also increasing the safety factor against main flange buckling.

Table 2: Factors of safety (FOS) for Iteration 1

Flexural Tension	Flexural Compression	Max Shear Stress	Glue Shear	Plate Bucking Case 1 Side Flange (K=4)	Plate Bucking Case 2 Main Flange (K=0.425)	Plate Bucking Case 3 web (K= 6)	Shear Buckling (K= 5)
4.193	1.484	2.294	5.273	23.265	3.417	15.611	3.118

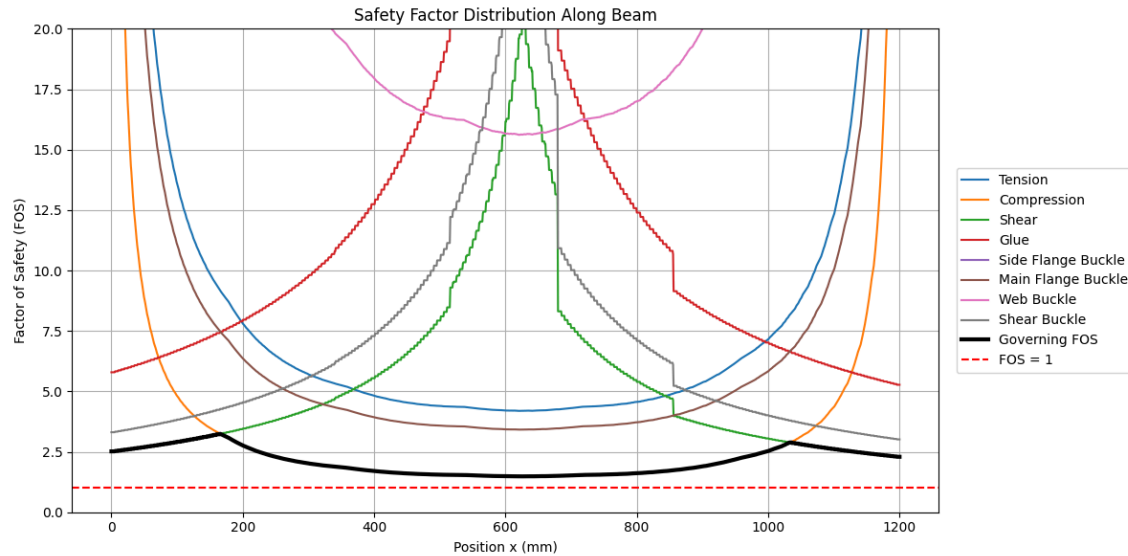


Figure 3: Graph of the Factor of Safety of 8 Different Failure Modes Against the Length of the Bridge

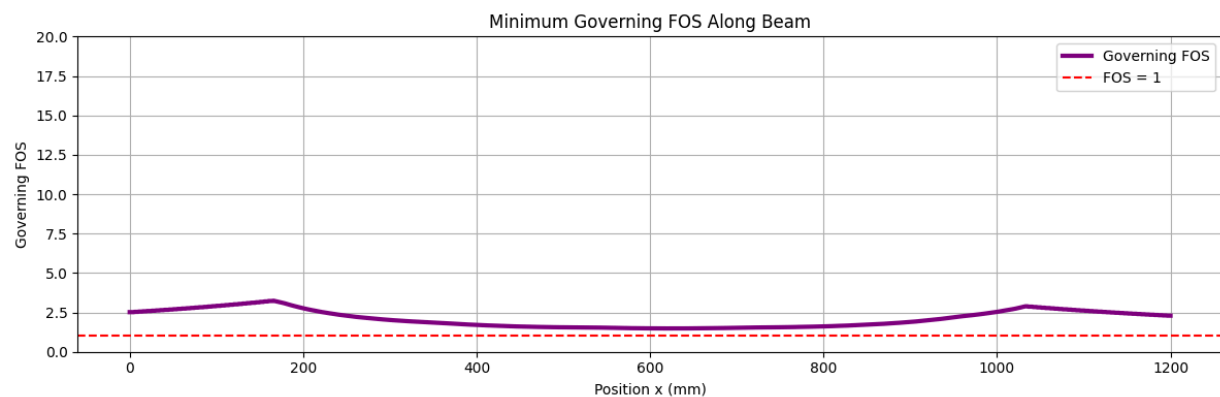


Figure 4: Graph of the Governing Factor of Safety (Flexural Compression) Against the Length of the Bridge (mm)

Minimum global FOS: 1.484 (flexural compression)

Estimated failure load (N): 593.468 N

Iteration #2 Increasing the Overall Web Height to 90 mm

After iteration 1, flexural compression became the lowest safety factor. In order to increase the safety factor, we knew we had to increase the I even more, and decided to do so by increasing the web height along the entire length of the bridge. This is because the total I is calculated with the formula $I = \Sigma A_i d_i^2 + I_i$ where the I of each rectangular part is computed via $\frac{bh^3}{12}$. Since the height is cubed, increasing the web height, being the dominant vertical dimension, provides the largest gain in I for alteration of any cross-section. The web height increased uniformly along the span from 75mm to 90mm. This change helped to increase the safety factor from

1.484 to 1.651, though it was not as significant as we had hoped. Despite the change, our current I value is not sufficient to counter high bending moments in the central region from 227.5mm to 827.5mm, as can be seen from Fig 6. This result guided out next criterion, which focused on increase of I near the midspan.

Table 3: Factors of safety (FOS) for Iteration 2

Flexural Tension	Flexural Compression	Max Shear Stress	Glue Shear	Plate Buckling Case 1 Side Flange (K=4)	Plate Buckling Case 2 Main Flange (K=0.425)	Plate Buckling Case 3 web (K= 6)	Shear Buckling (K= 5)
4.749	1.651	2.508	5.842	25.861	3.803	13.516	2.815

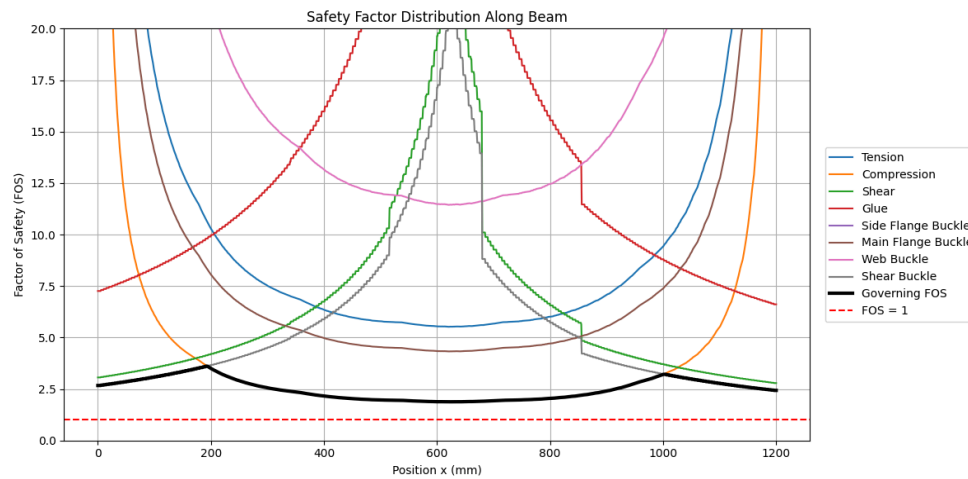


Figure 5: Graph of the Factor of Safety of 8 Different Failure Modes Against the Length of the Bridge After Iteration 2

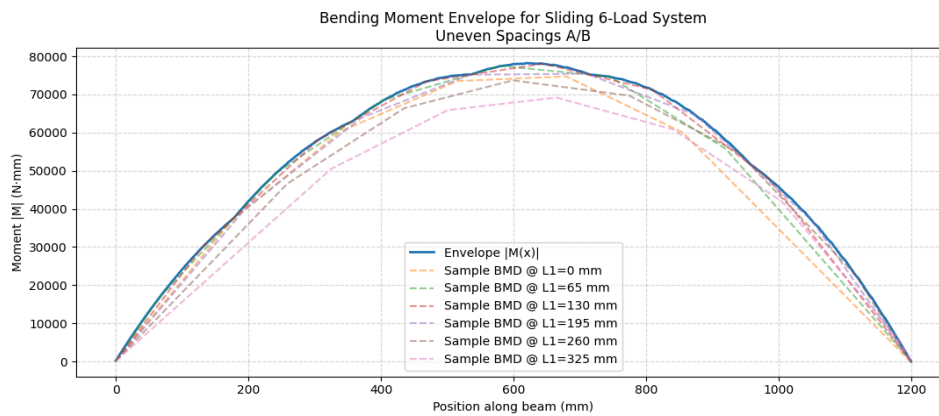


Figure 6: Graph of the Bending Moment Envelope for Load Case 2

Minimum global FOS: 1.651 (flexural compression)

Estimated failure load (N): 660.542 N

Iteration #3 Varying the Web Height

Although iteration 2 improved the flexural compression, it still remained the controlling failure mode.

Since flexural compression is given by: $\sigma = \frac{My}{I}$, further increasing I was necessary. However to conserve the material, we did not increase the web height across the entire length.

- ❖ From 0-297.5mm: the web height remained unchanged at 90mm
- ❖ Starting at 297.5 - 427.5mm: web height increased linearly to a maximum of 140mm, with an average height of 115mm across the slope.
- ❖ 427.5mm to 827.5mm the web height remained unchanged at height 140mm
- ❖ 827.5mm to 957.5: web height decreased linearly back to a height of 90mm, with an average height of 115mm across the slope.
- ❖ 957.5mm to 1255mm: the web height remained unchanged at 90mm

Since the bending moment grows significantly after the 200mm point, but again due to material constraint, we decided to begin the slope at the 270mm mark.

After this iteration our factor of safety against shear buckling plummeted. Shear stress is calculated with the

formula $\tau_{centroid} = \frac{V_{max} Q_{max}}{Ib}$. Increasing the I also helped to decrease shear stress along with compressive stress,

increasing the factor of safety against centroid shear. Despite this, increasing web height ended up making the bridge vulnerable to shear buckling in the middle. This is because shear buckling capacity is calculated by the

formula $\tau_{buckling} = \frac{5\pi^2 E}{12(1-u^2)} \left[\left(\frac{t}{diaphragm\ spacing} \right)^2 + \left(\frac{t}{web_{height} + glue\ tab_{height}} \right)^2 \right]$, and increasing web height

decreases the value of the $\left(\frac{t}{web_{height} + glue\ tab_{height}} \right)^2$ term, decreasing the overall capacity. Since the term with web

height is divided by the square of the web height, capacity for shear buckling decreases more than the shear stress does due to an increase in I. Thus, increasing web height made the bridge more susceptible to shear buckling in the centre, which guided our next design choice.

Table 4: Factors of safety (FOS) for Iteration 3

Flexural Tension	Flexural Buckling	Max Shear Stress	Glue Shear	Plate Buckling Case 1 Side Flange (K=4)	Plate Buckling Case 2 Main Flange (K=0.425)	Plate Buckling Case 3 web (K= 6)	Shear Buckling (K= 5)
9.961	3.097	4.157	10.748	48.502	7.133	6.401	1.600

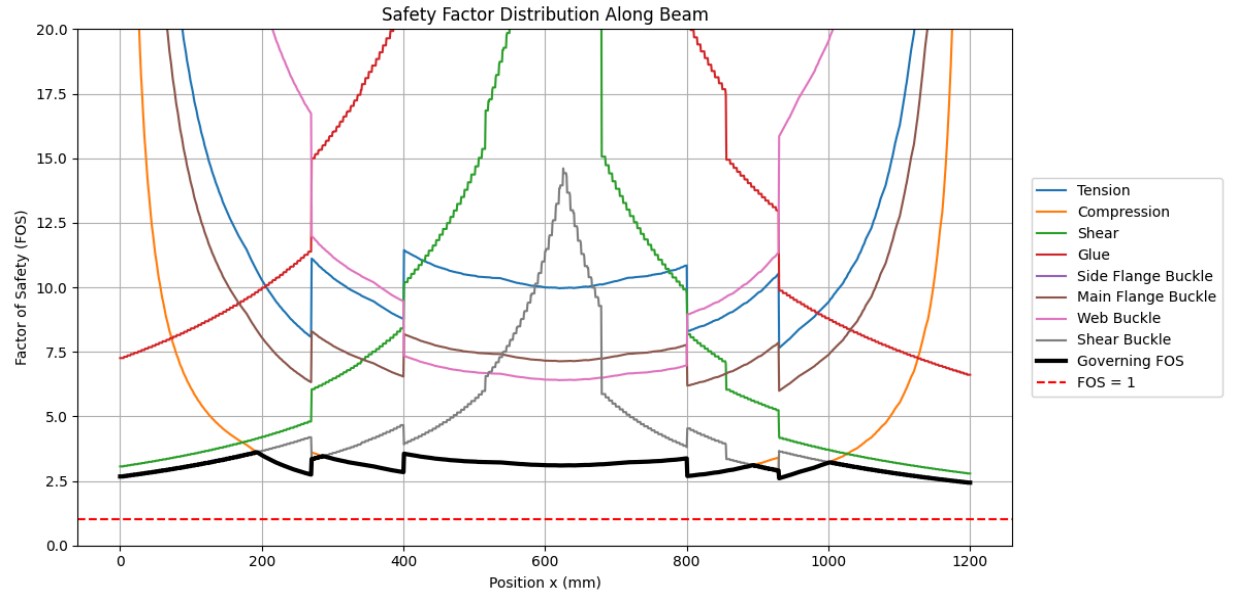


Figure 7: Graph of the Factor of Safety of 8 Different Failure Modes Against the Length of the Bridge After Iteration 3

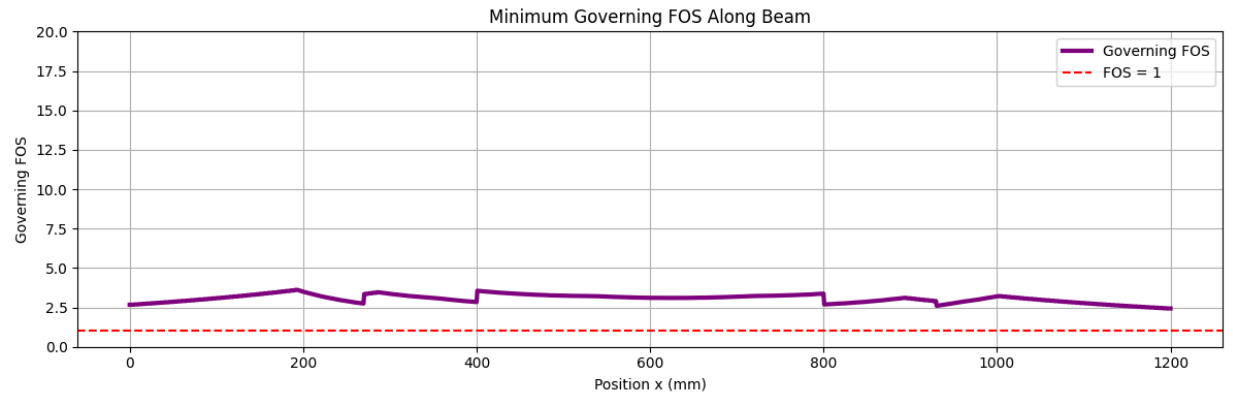


Figure 8: Graph of the Governing Factor of Safety (Shear Buckling) Against the Length of the Bridge (mm)

Minimum global FOS : 1.619 (Shear Buckling)

Estimated failure load (N): 647.511 N

Iteration #4 Adding Diaphragms and Decreasing Diaphragm Spacing

In the shear buckling capacity formula given as

$$\tau_{buckling} = \frac{5\pi^2 E}{12(1-u^2)} \left[\left(\frac{t}{diaphragm\ spacing} \right)^2 + \left(\frac{t}{web_{height} + glue\ tab_{height}} \right)^2 \right],$$

the only term we can modify with the given material properties is $\left[\left(\frac{t}{diaphragm\ spacing} \right)^2 + \left(\frac{t}{web_{height} + glue\ tab_{height}} \right)^2 \right]$. Since we did not want to

decrease our factor of safety values for flexural stresses and shear stress, we knew we had to modify the

$(\frac{t}{diaphragm\ spacing})^2$ rather than $(\frac{t}{web_{height} + glue\ tab_{height}})^2$. We decided to increase the number of diaphragms from 3 to 9, and change the spacing from uniform along the length of the bridge to uniform within specified regions (27.5:127.5, 127.5:297.5, 297.5:427.5, 427.5:827.5, 827.5:957.5, 957.5:1127.5, 1127.5:1227.5).

- ❖ From 27.5 to 127.5: two diaphragms spaced 33.3 mm apart
- ❖ From 127.5 to to 297.5: one diaphragm spaced 85mm from each bound
- ❖ From 297.5 to 427.5: one diaphragm spaced 130mm from each bound
- ❖ From 427.5 to 827.5: four diaphragm spaced 100mm apart
- ❖ From 827.5 to 957.5: two diaphragms spaced 33.3 mm apart
- ❖ From 957.5 to 1127.5: one diaphragm spaced 85mm from each bound
- ❖ From 1127.5to 1227.5: one diaphragm spaced 130mm from each bound

More diaphragms were allocated for the end regions (27.5-127.5mm) since centroid shear stress here is the strongest, and the central region (427.5-827.5mm) since the elevated web height in this region (maximum of 140mm) increases the risk of shear buckling the greatest. This allowed us to increase the shear buckling capacity, and since applied shear stress was constant, significantly increased the safety factor against shear buckling.

Table 5: Factors of safety (FOS) for Iteration 4

Flexural Tension	Flexural Compression	Max Shear Stress	Glue Shear	Plate Bucking Case 1 Side Flange (K=4)	Plate Bucking Case 2 Main Flange (K=0.425)	Plate Bucking Case 3 web (K= 6)	Shear Buckling (K= 5)
9,961	3.097	4.157	10.748	48.502	7.133	6.401	4.312

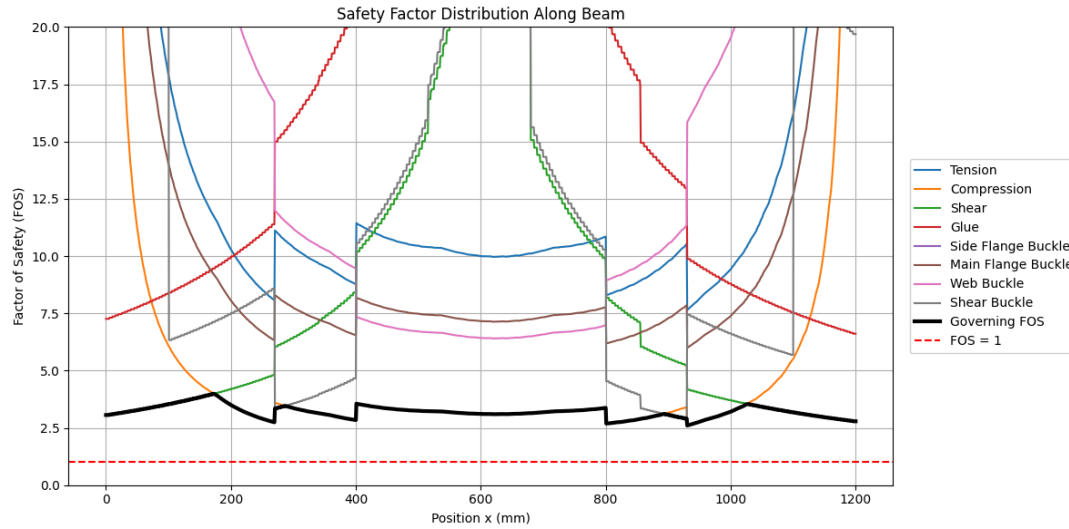


Figure 9: Graph of the Factor of Safety of 8 Different Failure Modes Against the Length of the Bridge After Iteration 4

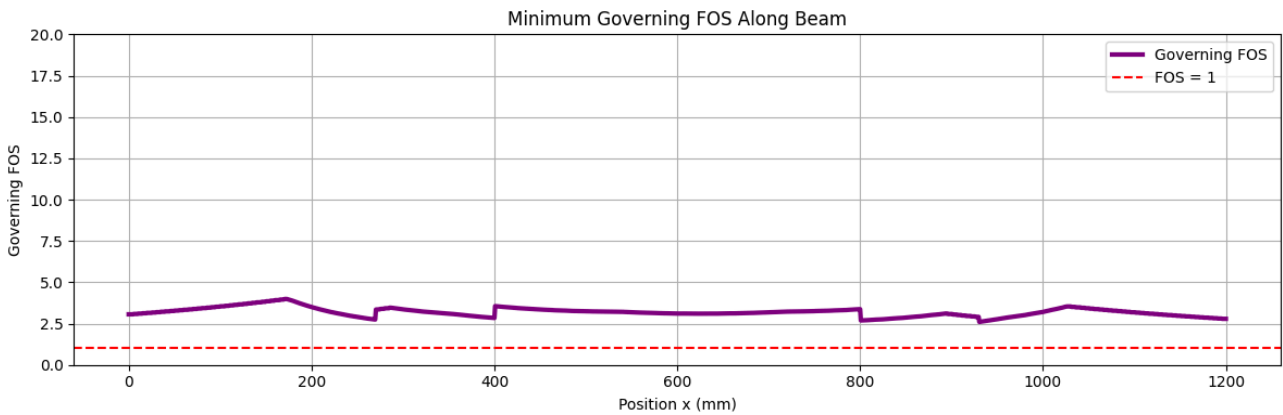


Figure 10: Graph of the Governing Factor of Safety (Shear Buckling) Against the Length of the Bridge (mm)

Minimum global FOS: 3.097 (flexural compression)

Estimated failure load (N): 1238.813 N

Iteration #5 Taking Out the Bottom on the Slope

While adding the diaphragms proved to improve shear buckling it also cost a lot of material. To stay within material constraints, we needed to cut down somewhere and finalized upon taking out the bottom deck from 297.5mm to 427.5 mm and 827.5mm to 957.5 since in these sloped sections neither shear nor bending moment are at maximum as seen in Fig 12. Therefore the bottom flange in these regions could be removed without compromising significantly flexural or shear capacity.

Table 6: Factors of safety (FOS) for Iteration 5

Flexural Tension	Flexural Compression	Max Shear Stress	Glue Shear	Plate Buckling Case 1 Side Flange (K=4)	Plate Buckling Case 2 Main Flange (K=0.425)	Plate Buckling Case 3 web (K= 6)	Shear Buckling (K= 5)
9,961	3.097	4.157	10.748	48.502	7.133	6.401	4.312

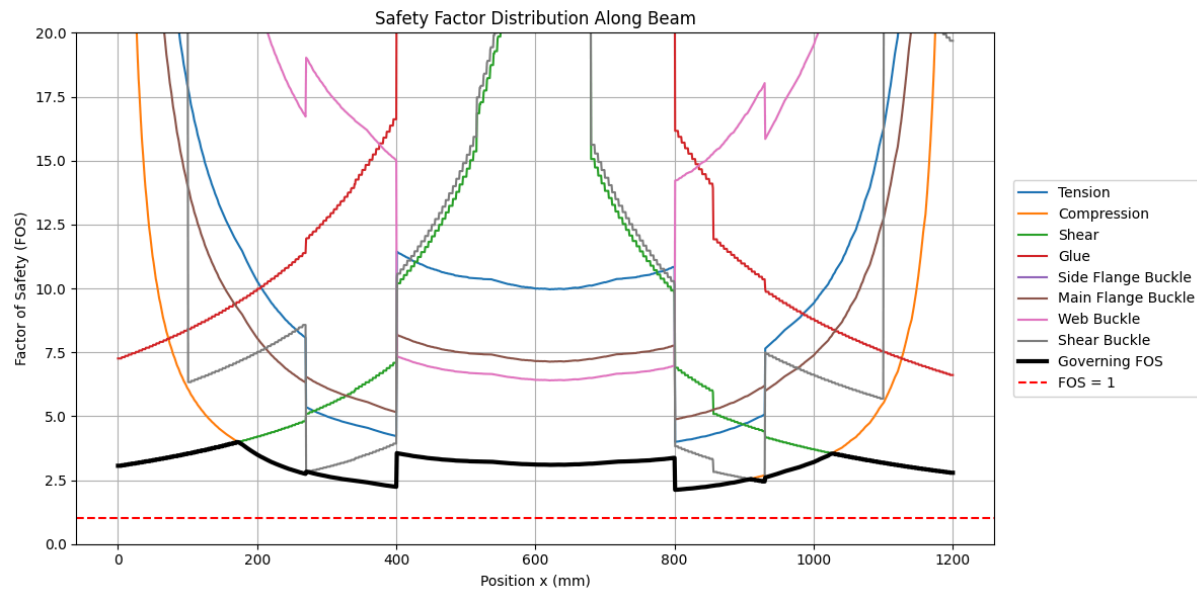


Figure 11: Graph of the Factor of Safety of 8 Different Failure Modes Against the Length of the Bridge After Iteration 5

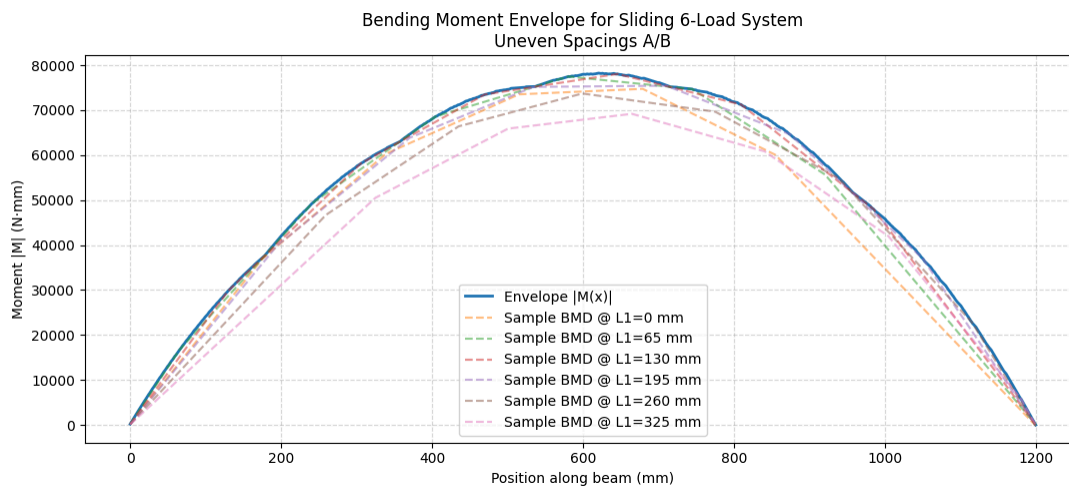


Figure 12: Graph of the Bending Moment Envelope for Load Case 2

Iteration 6 : Double Layering the Bottom Deck In the Middle

After removing the bottom flange on the sloped sections in iteration 5, we ended up with extra material, though not enough to have a bottom on both sides. To use this material more efficiently, it was used to reinforce our weakest factor of safety, flexural compression, by doubling the bottom layer from 530mm to 670mm, 70 mm on both sides of the centre of the bridge. This design choice increased the factor of safety against flexural compression without decreasing another factor. As it increased I reduced the flexural stress according to $\sigma = \frac{My}{I}$. It aided us to optimize the amount of material we had.

Table 7: Factors of safety (FOS) for Iteration 6

Flexural Tension	Flexural Compression	Max Shear Stress	Glue Shear	Plate Buckling Case 1 Side Flange (K=4)	Plate Buckling Case 2 Main Flange (K=0.425)	Plate Buckling Case 3 web (K= 6)	Shear Buckling (K= 5)
14.618	3.397	4.361	11.745	53.198	7.823	4.989	4.523

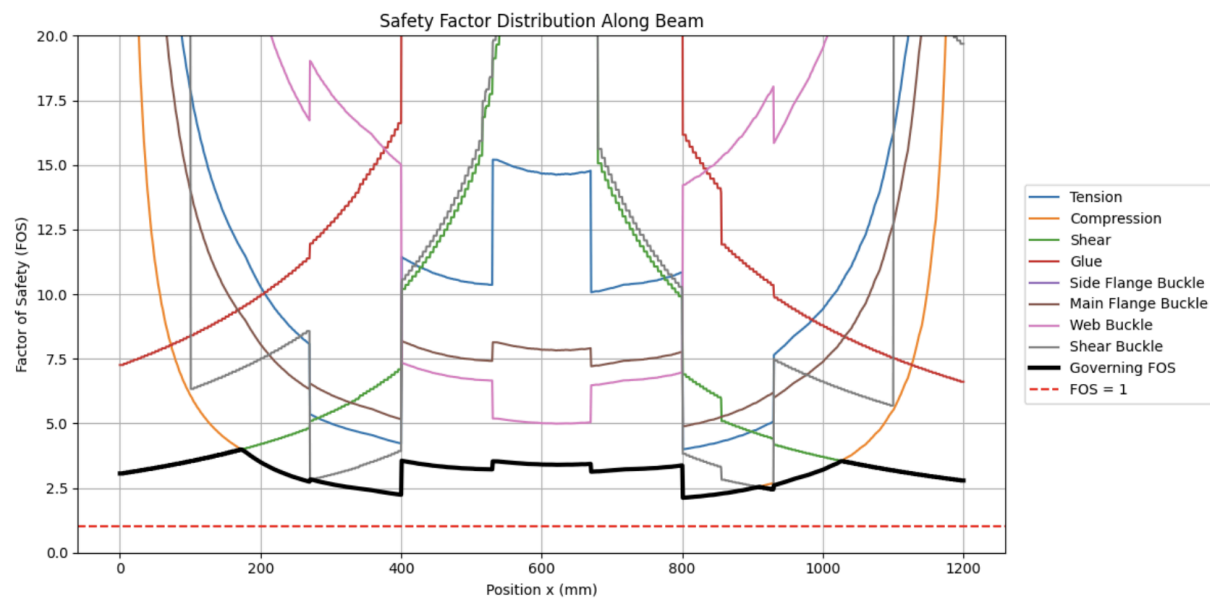


Figure 13: Graph of the Factor of Safety of 8 Different Failure Modes Against the Length of the Bridge After Iteration 6

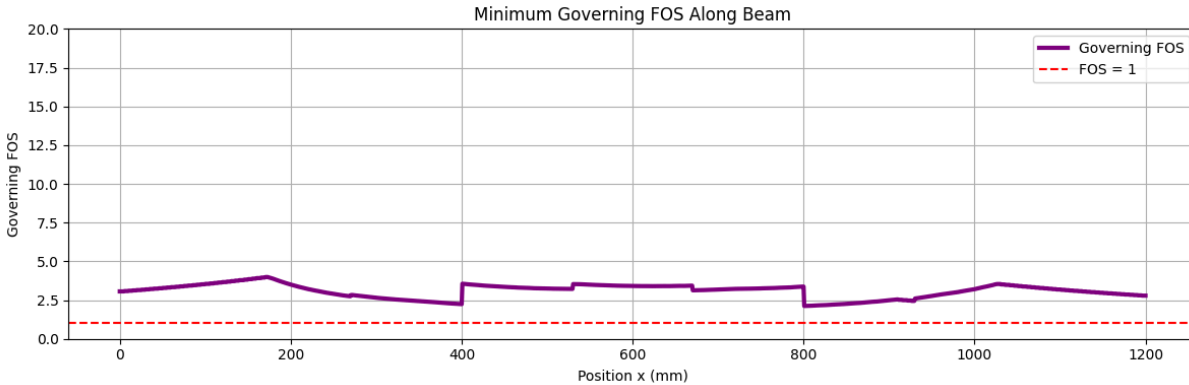


Figure 14: Graph of the Governing Factor of Safety (Shear Buckling) Against the Length of the Bridge (mm)

Minimum global FOS: 3.397(flexural compression)

Estimated failure load (N): 1358.759 N

Final Design and Conclusion

The final bridge design is the result of the six iterations:

- ❖ Adding a double layer over the entire length of the bridge
- ❖ Changing the web height over the entire length of the bridge
- ❖ Varying the web height from 90 to 140, between 297.5 to 957.5
- ❖ Adding and altering 11 diaphragm spacing
- ❖ Taking out bottom flange in sloped sections
- ❖ Introducing a double layered bottom deck in the middle

A theoretical load of 1358.799 is predicted from the final design, with the governing mode of failure due to material compression caused by flexural stress. In the end, compressive failure over shear buckling failure was favoured. This is because compressive failure tends to be more gradual and predictable than shear buckling, which results in sudden failure and complete loss in material strength (Chen & Lui, 2005). Furthermore, shear buckling is highly sensitive to geometric imperfections, which are inevitable during the construction process.

Despite the improved structural analysis shown by the theoretical calculations, the actual bridge is not expected to reach that load due to difficulties during the construction process. A major limitation was in the application of the glue. The cement glue was hard to apply uniformly, but more pressingly, it began to dry out while assembling and was not working at full capacity. This created regions of reduced strength between the layers, notably between the double-layered top deck and the webs. Since the theoretical failure load assumes perfect glue application, these errors can result in premature glue shear failure.

The quality of construction was further reduced by the misalignment of webs and diaphragms, caused by clamping pressure, measurement errors, and small shifts during handling. We found it hard to score the

matboard to create a clean fold, without deeply weakening the material, though scoring too lightly made it difficult to bend the piece at exactly 90°. Both errors introduced unintended curvature and soft spots along the web. Because plate buckling capacity depends on stiffness being perpendicular to the load, minor imperfection in the diaphragm's geometry can decrease the critical buckling stress. These construction imperfections make the structure more susceptible to premature shear buckling.

Although the theoretical failure load is 1358.759N, construction imperfection reduces the expected experimental failure load to approximately 40% of the original load to 543.5036 N. It is difficult to quantify to what extent construction errors affect the factors of safety against glue shear failure and shear buckling, but it is important to note that these modes can potentially override our predicted mode of failure: flexural compression.



Figures 15, 16 & 17: The completed and decorated bridge from side and top view and a picture of all the members holding it up (Left to right : Ezgi Muslu, Bice Vercelli, Zelal Arslan, Alex Meng).

References

Chen, W. F., & Lui, E. M. (Eds.). (2005). *Principles of Structural Design*. CRC Press.